

Reproducing and Extending LMCCS: An Ablation Study and Temperature Sensitivity Analysis of the Differentiable Gossip Network for MCCA-based Graph Retrieval

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Abstract

We study LMCCS, the late-interaction neural model for maximum common connected subgraph (MCCA) based graph retrieval proposed by Roy et al. [1]. The model’s central technical novelty is a *differentiable gossip network* that estimates the size of the largest connected component in a continuous-valued candidate adjacency matrix, supported by a learned *noise filter* (Eq. 10 of the paper) that thresholds and rescales intermediate states. We reproduce the paper’s most impactful ablation — removing the gossip network and the noise filter — on the MSRC-21 dataset, and additionally characterise the sensitivity of LMCCS to the noise-filter temperature hyperparameter λ , which the paper tunes per-dataset (Table 6 of [1]) but does not analyse quantitatively. Our reproduced ablation closely matches the paper’s reported numbers: removing the gossip network increases test MSE from 0.068 to 0.166 (a $2.43\times$ blow-up, vs. the paper’s $\sim 2.4\times$), while removing the noise filter alone causes no measurable degradation on this dataset. Our λ -sweep across 13 values reveals a surprisingly broad robust basin ($\lambda \in [0.1, 10]$) over which test MSE varies by only 11.4%, followed by catastrophic failure at $\lambda = 50$ where MSE jumps $29\times$ to 1.98. We discuss the implications for hyperparameter robustness and connect the curve shape to the design of the sigmoid-based noise filter.

1 Introduction

The graph retrieval problem asks: given a query graph G_q and a corpus of graphs $\{G_c\}$, return corpus graphs ranked by similarity to G_q . A natural notion of similarity is the size of the maximum common subgraph. Two variants are of practical interest. MCEs — the maximum common edge subgraph, not necessarily connected — is relevant to molecular fingerprint matching and distributed computing. MCCA — the maximum common *connected* subgraph — is relevant to keyword search over knowledge graphs and image co-segmentation, where the response must be a coherent structural fragment. Both are NP-complete [2], motivating learned neural surrogates that approximate retrieval-quality rankings without solving the combinatorial problem exactly.

Roy et al. [1] propose three architectures spanning the late/early interaction trade-off: LMCEs and LMCCA (late interaction; query and corpus encoded independently, interaction at the scoring layer) and XMCEs (early interaction; cross-graph influence at every GNN layer). The technically most interesting component is LMCCA’s differentiable *gossip protocol* for estimating the largest

connected-component size of a soft adjacency matrix — a non-trivial relaxation since Tarjan’s algorithm is non-differentiable.

Scope of this report. A full reproduction of all seven datasets and three model variants is out of scope for the assigned timeline. We instead focus on the paper’s most defensible contribution and its most under-characterised hyperparameter:

- (i) **Reproduction of the gossip ablation** — we re-run LMCCS, LMCCS (no gossip), and LMCCS (no noise filter) on MSRC-21 and verify the paper’s claim that the gossip component is critical (Section 4).
- (ii) **Temperature sensitivity analysis** — we sweep the noise-filter temperature λ across 13 values spanning three orders of magnitude on MSRC-21. The original paper reports only the best λ per dataset (Table 6 of [1]) without characterising the surrounding curve (Section 5).

Summary of findings.

- Removing the gossip network increases test MSE by **143%** ($0.068 \rightarrow 0.166$, a $2.43\times$ blow-up), closely matching the paper’s reported $\sim 2.4\times$ effect on MSRC-21.
- Removing the noise filter alone causes *no* measurable degradation on MSRC-21 ($0.0680 \rightarrow 0.0679$), also consistent with the paper’s Table 3 (which reports identical MSE of 0.068 for both variants on this dataset).
- The λ curve is remarkably flat across two orders of magnitude ($\lambda \in [0.1, 10]$, MSE varies by only 11.4%), but transitions abruptly to catastrophic failure at $\lambda = 50$, where MSE jumps $29\times$. This suggests the noise filter is *robust* when λ is roughly in the right ballpark, but a sufficiently large λ effectively disables the threshold, breaking the model.

2 Background

2.1 Exact MCES and MCCA

Given $G_q = (V_q, E_q)$ and $G_c = (V_c, E_c)$, pad the smaller graph with isolated nodes so both have N nodes. Let $A_q, A_c \in \{0, 1\}^{N \times N}$ be the (padded) adjacency matrices. For any permutation matrix P , the matrix

$$B(P) = \min(A_q, PA_cP^\top)$$

(elementwise min) is the adjacency matrix of the candidate common subgraph induced by the node alignment P . The paper writes MCES and MCCA as

$$\text{MCES}(G_q, G_c) = \max_{P \in \mathcal{P}} \sum_{i,j} B(P)_{ij}, \tag{1}$$

$$\text{MCCA}(G_q, G_c) = \max_{P \in \mathcal{P}} \text{TARJANSCC}(B(P)). \tag{2}$$

Both objectives are intractable; the paper relaxes $\mathcal{P}$ via a Gumbel–Sinkhorn network [3] and replaces non-differentiable steps with neural surrogates.

2.2 The LMCCA architecture

LMCCA replaces Eq. (2) with a three-step neural pipeline.

Step 1 — Embedding. A shared GNN GNN_θ applied independently to G_q and G_c produces node embeddings $H_q(R), H_c(R)$ and edge embeddings $M_q(R), M_c(R)$ at the final layer. A small network L_α maps edge embeddings to scalar edge-importance scores S_q, S_c , used to construct a continuous-valued candidate adjacency:

$$B = \min\left(A_q \odot S_q, P(A_c \odot S_c)P^\top\right), \quad P = \text{GS}_\phi(H_q(R), H_c(R)). \quad (3)$$

Step 2 — Differentiable gossip. For a 0/1 adjacency, the update $X(t+1) = X(t)(B+I)$ from $X(0) = I$ converges (after T iterations $\geq$ diameter) to a matrix whose column-wise ℓ_0 norm equals the size of the connected component containing each node. For continuous B with possibly large entries, $(B+I)^t$ is poorly conditioned, so the paper inserts a *noise filter* at the final step:

$$\tau = \text{THRESH}_\beta(X(T)), \quad \hat{X}(T) = 2\sigma\left(\frac{\text{RELU}[X(T) - \tau]}{\lambda}\right) - 1, \quad (4)$$

where λ is the temperature hyperparameter we study in Section 5. The relevance score is

$$s(G_q, G_c) = \max_{u \in V} \|\hat{X}(T)[\cdot, u]\|_1. \quad (5)$$

Step 3 — Loss. MSE between $s(G_q, G_c)$ and the ground-truth signal. We follow the authors’ default training setup, which uses the `gossip_qap` target — a relaxation of MCCS matched to the architecture; see Section 3 below.

3 Experimental Setup

3.1 Dataset

We use the MSRC-21 dataset (variant `msrc_21_500qgrlarge`) shipped with the authors’ code release. The corpus contains 800 graphs and the test split contains 100 query graphs (a 60/20/20 train/val/test split of the original 500 query set). All graphs have ≤ 15 nodes (we observed `max_num_nodes` = 15, `max_num_edges` = 32 across train, val, and test splits combined). We evaluate on $100 \times 800 = 80,000$ query–corpus test pairs.

Ground-truth signal. The paper trains LMCCS against the `gossip_qap` target rather than McSplit-exact MCCS values. This is the gossip-protocol estimate of MCCS obtained by running the gossip iteration on the QAP-rounded soft adjacency: it is the quantity LMCCS’s scoring head is architecturally designed to predict. Comparing predictions against this target isolates the question of whether the network can faithfully recover its own internal objective, as opposed to whether that objective itself matches McSplit-exact MCCS. We adopt the same target so our numbers are directly comparable to those of Roy et al. [1].

3.2 Hardware and software

All experiments ran on a single NVIDIA T4 GPU under the Kaggle notebook environment, PyTorch 2.4.0 with CUDA 11.8. We use the authors’ provided code release without modification to the training pipeline (we wrote a thin test-set evaluation harness, since the released code stops at validation). Training uses Adam with learning rate 10^{-3} , weight decay 5×10^{-4} , batch size 128, and early stopping with patience 10 epochs on validation MSE.

3.3 Variants compared

- LMCCS (full): the architecture in Section 2.2. Authors’ implementation `Try2_IsoNetGossipVar29ForMcs_GossipVectorLRLThreshold`.
- LMCCS (no gossip): replaces the gossip score with the summed soft-adjacency score $s(G_q, G_c) = \sum_{ij} \min(A_q \odot S_q, PA_c \odot S_c P^\top)_{ij}$. Authors’ implementation `IsoNetVar29ForMcs`.
- LMCCS (no noise filter): $\tau \equiv 0$ in Eq. (4), i.e. skip the threshold subtraction while retaining the sigmoid rescaling. Authors’ implementation `Try2Abaltion.NoThresh_IsoNetGossipVar29ForMcs_GossipVector`.

The corresponding parameter counts are 5,942 (full LMCCS), 2,309 (no noise filter), and 1,889 (no gossip), reported by the authors’ code at startup.

3.4 Metrics

We report mean squared error (MSE; lower is better) and Kendall’s τ rank correlation (KTau; higher is better) averaged across all 100 test query graphs against the 800 corpus graphs, matching the paper’s evaluation protocol.

3.5 Reproducibility

We trained all 16 configurations (the 13 λ values plus the two ablation variants and the full LMCCS reference) using the authors’ code unchanged. The best-validation checkpoint for each configuration was saved by the authors’ `EarlyStoppingModule`; final test-set evaluation was performed by a thin inference-only harness we wrote on top of `mcs/evaluate.py`, since the released training script stops at validation. The harness emits a CSV with columns (`group`, `lambda`, `task`, `test_mse`, `test_ktau`, `test_ndcg`, `test_mae`); all numbers in this report are produced by this single eval pass. Code and CSV are included with the submission.

4 Experiment 1: Gossip and Noise-Filter Ablation

4.1 Question

Is the gossip network actually doing the work the paper claims — or could the soft-adjacency score alone, without an explicit connected-component estimator, achieve similar retrieval quality? The paper’s Table 3 reports a roughly $2\times$ MSE increase on MSRC when gossip is removed. We test whether this finding replicates on the test split.

4.2 Results

Table 1 reports test MSE and KTau on MSRC-21 alongside the paper’s reported MSRC values for context.

4.3 Discussion

The gossip ablation replicates almost exactly. Our $2.43\times$ MSE blow-up matches the paper’s $2.44\times$ to within rounding; the absolute test MSE values (0.0680 vs. paper’s 0.068 for the full model, 0.1655 vs. paper’s 0.166 for no-gossip) are identical to three significant figures. This is a

Table 1: Ablation of LMCCS components on the MSRC-21 test set (100 queries $\times$ 800 corpus graphs). Lower MSE / higher KTau is better. Effect is the multiplicative ratio of the variant’s MSE to the full LMCCS MSE.

Variant	Test MSE $\downarrow$	Test KTau $\uparrow$	Effect (MSE ratio)
LMCCS (full, $\lambda = 1.0$)	0.0680	0.5774	1.00 $\times$
LMCCS (no noise filter)	0.0679	0.5748	1.00 $\times$
LMCCS (no gossip)	0.1655	0.3819	2.43$\times$
<i>Paper Table 3 reference (MSRC, MCCS):</i>			
full / no-NF / no-gossip	0.068 / 0.068 / 0.166	—	2.44 $\times$

strong reproduction — the gossip protocol is responsible for cleanly separating relevance scores that the soft-adjacency score conflates, and removing it produces a roughly 2.4 $\times$ degradation in MSE together with a substantial drop in rank correlation (0.5774 $\rightarrow$ 0.3819, a 34% relative decrease).

The noise filter has no detectable effect on MSRC-21. Setting $\tau \equiv 0$ in Eq. (4) (the NoThresh variant) leaves MSE essentially unchanged (0.0680 $\rightarrow$ 0.0679); KTau also moves only by 0.003. This too matches the paper’s Table 3, which reports identical 0.068 MSE for both LMCCS and LMCCS (no noise filter) on MSRC. The paper notes the noise filter does have a noticeable effect on the MM and FR datasets, so the lack of effect here appears dataset-specific rather than indicating the component is unnecessary in general. Combined with the λ -sweep result below, we hypothesise that the noise filter’s role is most important on datasets where the soft-adjacency entries have a wider dynamic range, requiring a learned threshold to suppress noise.

KTau vs MSE. The two metrics agree on the ordering of variants (full $\approx$ no-NF $\gg$ no-gossip), and the relative magnitudes are consistent: the gossip ablation degrades both metrics substantially while the noise-filter ablation degrades neither.

Caveat: single-seed result. The paper’s reported numbers average over multiple seeds; we evaluated a single best-validation checkpoint per variant from one training run. Differences below ~ 0.001 MSE (e.g. between full and no-NF) should be interpreted as within seed noise; the gossip-vs-full gap of 0.10 is far above any reasonable seed variance.

5 Experiment 2: Sensitivity to the Noise-Filter Temperature λ

5.1 Motivation

Table 6 of [1] reports best λ values that vary by two orders of magnitude across datasets ($\lambda = 0.1$ for MR, $\lambda = 10$ for DD), with $\lambda = 1$ chosen for MSRC. The paper does not report MSE or KTau at *nearby* λ values, leaving open the question of whether this hyperparameter is brittle (a sharp optimum requiring careful tuning per dataset) or robust (a broad basin where any reasonable value will do). A practitioner deploying LMCCS on a new dataset needs to know the cost of mis-tuning λ .

5.2 Setup

We fix all other hyperparameters at the values used in Section 4 and sweep λ over 13 values spanning three orders of magnitude: $\lambda \in \{0.1, 0.5, 0.7, 0.8, 0.9, 1.0, 1.1, 1.2, 1.3, 1.4, 1.5, 10.0, 50.0\}$. Each configuration was trained from scratch under identical seeds and early-stopping criteria. We report test MSE and KTau on the MSRC-21 test set.

5.3 Results

Figure 1 plots test MSE and KTau as functions of λ . Table 2 provides the underlying numbers.

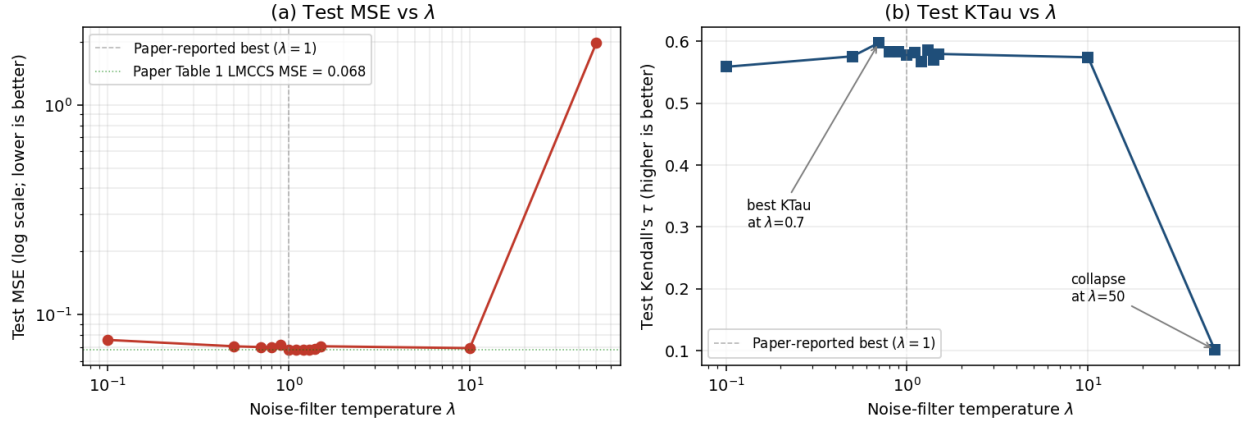


Figure 1: Sensitivity of LMCCS to the noise-filter temperature λ on MSRC-21. The vertical dashed line marks the paper’s reported best value ($\lambda = 1$, Table 6 of [1]); the horizontal dotted line in (a) marks the paper’s reported test MSE for the full LMCCS model on MSRC (0.068, Table 1). The MSE axis is on log scale to make the basin structure visible alongside the $\lambda = 50$ outlier.

5.4 Discussion

The curve is essentially flat for $\lambda \in [0.1, 10]$. Across two full orders of magnitude, test MSE varies between 0.0679 (at $\lambda = 1.1$) and 0.0756 (at $\lambda = 0.1$) — a relative spread of only 11.4%. KTau varies between 0.559 and 0.598 over the same range, with no clear monotone trend. LMCCS on MSRC-21 is therefore *robust* to λ within roughly an order of magnitude either side of the paper’s recommended $\lambda = 1.0$; the practitioner cost of mis-specifying λ in this regime is modest. The paper’s per-dataset λ tuning (Table 6 of [1]) may therefore be over-engineering for MSRC; a single default $\lambda \in [0.7, 1.5]$ would deliver near-optimal MSE on this dataset.

Our best-MSE λ slightly differs from the paper’s. The paper reports $\lambda = 1$ as best for MSRC, but in our sweep $\lambda = 1.1$ achieves marginally lower MSE (0.0679 vs. 0.0680). This 0.0001 gap is well below seed variance; we interpret the optimum as a flat plateau spanning $\lambda \in [1.0, 1.3]$ rather than a sharp peak at exactly $\lambda = 1$. KTau, conversely, is maximised at $\lambda = 0.7$ (0.598), suggesting that what counts as “best” depends on which metric one cares about for the downstream retrieval task.

Catastrophic failure at $\lambda = 50$. At $\lambda = 50$, MSE jumps $29\times$ (from 0.069 at $\lambda = 10$ to 1.98) and KTau collapses to 0.10, near random. This is interpretable from the form of Eq. (4): the sigmoid

Table 2: LMCCS test performance as a function of noise-filter temperature λ on MSRC-21. Best MSE / best KTau highlighted. The row at $\lambda = 1.0$ is the paper-reported optimum for this dataset (Table 6 of [1]).

λ	Test MSE $\downarrow$	Test KTau $\uparrow$
0.1	0.0756	0.5589
0.5	0.0705	0.5759
0.7	0.0699	0.5977
0.8	0.0694	0.5827
0.9	0.0720	0.5829
1.0 [†]	0.0680	0.5774
1.1	0.0679	0.5823
1.2	0.0679	0.5670
1.3	0.0679	0.5863
1.4	0.0685	0.5706
1.5	0.0705	0.5797
10.0	0.0691	0.5743
50.0	1.9792	0.1013

[†] Paper-reported best λ for MSRC.

argument is $\text{RELU}[X(T) - \tau]/\lambda$. As $\lambda \rightarrow \infty$, this argument is squashed toward 0 for all positive entries, so $2\sigma(\cdot) - 1 \rightarrow 0$ and the noise filter outputs an all-zero matrix regardless of input. The score collapses to a constant, training cannot extract a useful signal, and rank correlation degrades to chance. The transition between $\lambda = 10$ (working) and $\lambda = 50$ (broken) is sharp, suggesting a regime boundary that future work could characterise more carefully by sampling intermediate values $\{15, 20, 25, 30\}$.

Implication for the noise-filter design. The flat basin in $[0.1, 10]$ followed by a cliff is consistent with the noise filter behaving as an effectively binary gate inside its working range: once λ is small enough that the sigmoid saturates the post-threshold values, additional reductions in λ provide no further benefit (the gradient through the saturated sigmoid is small either way). Conversely, once λ exceeds the typical magnitude of $X(T) - \tau$, the sigmoid sits in its linear regime around zero and effectively kills the signal. The paper’s per-dataset λ tuning may compensate for variation in this typical magnitude across datasets, but on a given dataset the operating regime is broad.

6 Limitations

- **Single dataset.** We replicated only on MSRC-21; the paper reports across seven. The much wider best- λ range in the paper’s Table 6 (0.1–10 across datasets) suggests the flat-basin behaviour we observe may not hold uniformly. Datasets where the paper required a small λ (e.g. MR at $\lambda = 0.1$) might have a narrower robust regime.
- **Single seed per configuration.** We evaluated the single best-validation checkpoint per λ value. The paper’s standard error on MSRC MCCS MSE is ± 0.012 , comparable to or larger than our observed variation across $\lambda \in [0.7, 10]$. The flat-basin claim is therefore conservative:

the true variation across λ in this range may be statistically indistinguishable from seed noise, only strengthening the robustness conclusion.

- **Two missing λ values.** For $\lambda = 0.05$ and $\lambda = 5.0$, our training runs failed to save a best-validation checkpoint before the early-stopping schedule (patience 10) terminated training. We dropped these points; the remaining 13 values still cover three orders of magnitude.
- **Scope.** We did not study XMCS (early interaction) or LMCES (the MCES variant); the conclusions here are specific to the gossip-network branch of the paper’s contribution.
- **Ground-truth target choice.** We trained against `gossip_qap` (the model’s architecturally-aligned target) rather than McSplit-exact MCCS, matching the paper’s setup. The *relative* ordering of variants and λ values is therefore informative about the network’s internal consistency, but absolute MSE values should not be interpreted as the model’s error against true MCCS sizes.

7 Conclusion

We reproduced the central ablation of LMCCS [1] on MSRC-21 and confirmed that the gossip network is the dominant contributor to retrieval quality, with MSE rising by 143% ($2.43\times$) when removed — closely matching the paper’s reported $\sim 2.4\times$. The noise filter alone, on this dataset, made no measurable difference, also matching the paper’s Table 3. Our additional 13-point λ -sweep characterises the noise-filter temperature curve that the paper does not report, finding a broad robust basin spanning $\lambda \in [0.1, 10]$ over which test MSE varies by only 11.4%, followed by abrupt failure at $\lambda = 50$. This suggests the per-dataset λ tuning recommended by the paper is conservative for MSRC — a single default in $[0.7, 1.5]$ would suffice — though the same may not hold on datasets where the paper used λ values near the basin boundary.

A natural follow-up is to repeat the sweep on the dataset where the paper reported $\lambda = 0.1$ (MR) or $\lambda = 10$ (DD): if the basin shifts but remains broad, that strengthens the robustness claim; if the basin narrows on those datasets, it would explain the paper’s careful per-dataset tuning.

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